

Why this note exists

9999-density-trans.qmd works out the density-transformation machinery MSL needs: a C^1 bijection Φ , its Jacobian $J_i(\varepsilon)$, and a 1-D integral over the one remaining unobservable ε . This note works out the alternative to all of that — **ELVIS** (Schennach 2014, *Econometrica*, “Entropic Latent Variable Integration via Simulation,” @Schennach2014) — at the same level of derivation detail: not just citing the result, but rederiving where the tilt comes from, and then applying it, step by step, first to stage 2 alone and then to the fully unified single-latent model.

The starting point is a failure, not the theorem. Working through *why* a naive Monte-Carlo GMM moment doesn’t work is what motivates every piece of ELVIS’s machinery below — so it’s worth restating precisely before introducing the fix.

The bug ELVIS is built to fix

Stage 2’s evasion FOC, solved for ψ and written as a function of the one unobservable ε (holding firm it ’s own data $(M_{it}^*, \mathcal{V}_{it}, \tilde{\mathcal{W}}_{it})$ fixed), contains a nonlinear term

$$\ell_{it}(\varepsilon; \lambda) \equiv \ln(1 - 2\lambda M_{it}^* (1 - e^{-\mathcal{V}_{it} - \varepsilon})).$$

ε_{it} is not observed. The naive fix is to simulate $\varepsilon^{(s)} \sim \hat{f}_\varepsilon$, holding the firm’s own data fixed, and average:

$$\hat{\ell}_{it}^S(\lambda) = \frac{1}{S} \sum_{s=1}^S \ell_{it}(\varepsilon^{(s)}; \lambda) \xrightarrow{a.s.} \bar{\ell}_{it}(\lambda) \equiv \int \ell_{it}(\varepsilon; \lambda) f_\varepsilon(\varepsilon) d\varepsilon.$$

This is a specific, computable number. It is also **the wrong number**: the moment condition needs firm it ’s value at its own realized ε_{it} , not the unconditional average over every value ε could have taken. Because ℓ_{it} is nonlinear in ε ,

$$\bar{\ell}_{it}(\lambda) \neq \ell_{it}(\varepsilon_{it}; \lambda) \quad \text{generically.}$$

Write $\eta_{it}(\lambda) \equiv \ell_{it}(\varepsilon_{it}; \lambda) - \bar{\ell}_{it}(\lambda)$. Swapping in $\bar{\ell}_{it}(\lambda)$ silently injects $-\eta_{it}(\lambda)$ into every firm’s moment — a nonlinear function of the firm’s own ε_{it} , entangled with $(M_{it}^*, \mathcal{V}_{it})$, so it cannot be argued away by $\varepsilon \perp Z$ alone: for Z to be *relevant* for λ it must correlate with the firm’s evasion behavior, and that is exactly what $\eta_{it}(\lambda)$ also depends on. Recasting the simulation in M instead of ε does not help — $M_{it}^{(s)} = \beta P_t Y_{it} / \rho_t \cdot e^{-\varepsilon^{(s)}}$ and $e_{it}^{(s)} = M_{it}^* - M_{it}^{(s)}$ reconstruct the *identical* map, since $e^{-\mathcal{V}_{it}} = \beta P_t Y_{it} / (\rho_t M_{it}^*)$. Same nonlinearity, same missing reweighting.

What’s missing, named precisely: the correct object is not the plain average over $f_\varepsilon(\varepsilon)$, but the average over the *posterior* distribution of ε implied by having observed this particular firm’s data — i.e. some reweighting of f_ε by how consistent each candidate ε is with the firm’s own outcome. MSL supplies exactly that reweighting (that’s what $f_\psi(\psi(\varepsilon))J_i(\varepsilon)f_\omega(\omega(\varepsilon))$ is doing inside the likelihood integral), but MSL needs f_ψ to do it, which is precisely the object stage 2 is trying to recover — circular, if the goal is a moment-only estimator that doesn’t first assume a parametric f_ψ . ELVIS’s contribution is to derive the correct reweighting **without ever specifying** f_ψ — solving for it directly from the moment condition itself.

Schennach’s Theorem 2.1

Setup

Let U be an unobserved random vector, Z an observed vector with (data-identified) marginal π , and $g(u, z, \theta)$ a known, possibly nonlinear/nonmonotone/nonseparable function into $\mathbb{R}^{d_g}$. Consider the question: does there exist θ_0 and *some* conditional distribution $\mu(u | z)$ of U given Z — consistent with whatever support restrictions the model imposes — such that

$$E_{\mu \times \pi}[g(U, Z, \theta_0)] = 0? \quad (1)$$

This is an infinite-dimensional question: $\mu(\cdot | z)$ ranges over an entire space of conditional distributions, one for (almost) every value of z . Directly searching over that space is not implementable.

The result

Fix **any** dominating conditional measure $\rho(u | z)$ whose support contains the support the model implies for U given Z (the *support* is a modeling choice; the *shape* of ρ is not — see Remark 2.3 below). Define the **tilted moment**

$$\tilde{g}(z, \theta, \gamma) \equiv \frac{\int g(u, z, \theta) \exp(\gamma' g(u, z, \theta)) d\rho(u | z)}{\int \exp(\gamma' g(u, z, \theta)) d\rho(u | z)}. \quad (2)$$

Schennach’s Theorem 2.1: Equation 1 holds **if and only if** there exists (θ_0, γ_0) solving the finite-dimensional system

$$E_\pi[\tilde{g}(Z, \theta_0, \gamma_0)] = 0. \quad (3)$$

An infinite-dimensional existence question about conditional distributions collapses to a finite-dimensional system in $(\theta, \gamma) \in \mathbb{R}^{\dim \theta} \times \mathbb{R}^{d_g}$ — the same dimension count as an ordinary GMM problem, just with d_g extra parameters γ (one per moment). The “only if” direction is the nontrivial half of her proof (not reproduced here); the “if” direction is the easier one and is exactly what the derivation below makes concrete.

Where the tilt comes from — an entropy-maximizing projection

The “if” direction, worked out by hand: fix θ and z , and ask whether *some* $\mu(\cdot | z)$ satisfying $E_\mu[g(U, z, \theta)] = 0$ exists. Among all such μ , consider the one closest to the chosen dominating measure $\rho(\cdot | z)$ in Kullback–Leibler divergence — i.e. solve

$$\min_{\mu} \int \mu(u) \ln \frac{\mu(u)}{\rho(u)} du \quad \text{s.t.} \quad \int g(u, z, \theta) \mu(u) du = 0, \quad \int \mu(u) du = 1.$$

Lagrangian (multipliers $\gamma \in \mathbb{R}^{d_g}$ on the moment constraint, ν on the adding-up constraint):

$$\mathcal{L} = \int \mu(u) \ln \frac{\mu(u)}{\rho(u)} du - \gamma' \int g(u, z, \theta) \mu(u) du - \nu \left(\int \mu(u) du - 1 \right).$$

Pointwise first-order condition, $\partial \mathcal{L} / \partial \mu(u) = 0$:

$$\ln \frac{\mu(u)}{\rho(u)} + 1 - \gamma' g(u, z, \theta) - \nu = 0 \quad \implies \quad \mu(u) = \rho(u) \exp(\gamma' g(u, z, \theta) + \nu - 1).$$

Now pin down ν by normalizing. The factor $\exp(\nu - 1)$ in the FOC’s solution does not depend on u — write it as a constant $C(z, \theta, \gamma) \equiv \exp(\nu - 1)$, so the solution reads $\mu(u) = C \cdot \rho(u) \exp(\gamma' g(u, z, \theta))$. Impose $\int \mu(u) du = 1$:

$$1 = \int \mu(u) du = C \int \rho(u) \exp(\gamma' g(u, z, \theta)) du \quad \implies \quad C = \left[\int \rho(u') \exp(\gamma' g(u', z, \theta)) du' \right]^{-1}.$$

This determines C (equivalently ν) as a function of (z, θ, γ) only — never of u , since it comes from integrating u out. Substituting this C back into $\mu(u) = C \cdot \rho(u) \exp(\gamma' g(u, z, \theta))$ delivers the closed form

$$\mu^*(u | z; \theta, \gamma) = \frac{\rho(u | z) \exp(\gamma' g(u, z, \theta))}{\int \rho(u' | z) \exp(\gamma' g(u', z, \theta)) du'}. \quad (4)$$

This is the entropy-maximizing (equivalently, KL-closest-to- ρ) member of the family of tilts of ρ . Substituting μ^* into $E_{\mu^*}[g(U, z, \theta)] = 0$ reproduces Equation 2 exactly — $\tilde{g}(z, \theta, \gamma)$ is $E_{\mu^*}[g]$. So γ solving $\tilde{g}(z, \theta, \gamma) = 0$ is precisely the multiplier that makes the exponentially-tilted ρ satisfy the moment condition. This is the **missing reweighting** identified in the previous section — not assumed, not guessed, but *solved for* as a Lagrange multiplier on the moment constraint. That solved-for γ is what the naive Monte-Carlo average silently set to zero (an untilted ρ , i.e. $\gamma = 0$, is exactly $\bar{\ell}_{it}(\lambda)$ from the bug above).

What Theorem 2.1 adds beyond this derivation is the claim that restricting attention to the *entropy-maximizing* μ^* loses nothing relative to searching over *all* candidate μ : if some (any) μ satisfies the moment condition, so does the entropy-maximizing one, for the right γ . That equivalence — not just “one sufficient construction” but “this is not a restriction” — is the theorem’s real content, and it is what licenses working entirely in (θ, γ) -space, finite-dimensional, without loss.

Remark 2.3 — the dominating measure ρ doesn’t matter

Equation 1 never mentions ρ at all — it’s a question about whether *some* μ exists. Theorem 2.1 says this is equivalent to solvability of Equation 3 *for any valid choice* of ρ (any ρ whose support covers the model-implied support of $U \mid Z$). Different choices of ρ reparametrize the *same* underlying family of reachable tilts μ^* , so the value of θ_0 that makes Equation 1 true cannot depend on which ρ was used to search for it. Consequence for this project: **the choice of ρ only ever affects simulation efficiency (how much variance the ratio-of-averages estimator below has), never the point estimate itself.** Concretely, reusing stage 1’s $\hat{f}_\varepsilon$ (translated into a distribution over M) as ρ is a legitimate, efficiency-motivated choice — but nothing is broken by using something cruder (e.g. uniform on the feasible support), only slower convergence.

Applying ELVIS to stage 2 alone: recasting the evasion moment in M

The one unobservable

Take M_{it} — the true, unobserved materials level — as U , rather than ε_{it} . This mirrors Schennach’s own canonical errors-in-variables example (her Example 1.5: true regressor X^* , observed $X = X^* + \Delta X$) structurally exactly: $M_{it}^* = M_{it} + e_{it}$ is the observed, contaminated version of the true M_{it} .

Given M_{it} , every other quantity in the stage-2 moment is a deterministic function of data and θ :

$$e(M) = M_{it}^* - M, \quad \varepsilon(M) = \ln\left(\frac{M_{it}^*}{M}\right) - \mathcal{V}_{it}, \quad \omega(M) = \tilde{\mathcal{W}}_{it} - (1 - \beta)\varepsilon(M).$$

The first follows from the definition of evasion; the second by solving **?@eq-eva-emp** (i.e. $\mathcal{V}_{it} = \ln(M_{it}^*/M_{it}) - \varepsilon_{it}$) for ε ; the third from **?@eq-tilde-big-w** solved for ω . The evasion-FOC moment, written directly as a function of M :

$$g(M, Z_{it}, \theta) \equiv \psi(M) = h(e(M)) - \delta_0 + \delta_1 \omega(M) - \delta_2 \omega(M)^2. \quad (5)$$

Recasting in M changes *which variable ELVIS treats as unobserved* — it is not a different moment from the one **?@eq-foc-psi** already defines.

Support

The FOC's own feasibility ceiling, $e_{it} \in (0, \frac{1}{2\lambda})$, becomes the support restriction on M directly:

$$e(M) = M_{it}^* - M < \frac{1}{2\lambda} \iff M > M_{it}^* - \frac{1}{2\lambda}, \quad e(M) > 0 \iff M < M_{it}^*,$$

so $\mathcal{U}_{Z,\theta} = (M_{it}^* - \frac{1}{2\lambda}, M_{it}^*)$ — a genuinely θ -dependent support (it moves with λ). Schennach's framework explicitly accommodates θ -dependent support (her §4.1); it is not a complication here, it's the domain the model already hands over for free.

The estimator

Simulate $M^{(s)} \sim \rho(\cdot | Z_{it})$ (e.g. via Metropolis, or directly if ρ is chosen tractable) and form the sample analog of Equation 2:

$$\hat{g}(Z_{it}, \theta, \gamma) = \frac{\frac{1}{S} \sum_{s=1}^S g(M^{(s)}, Z_{it}, \theta) \exp(\gamma' g(M^{(s)}, Z_{it}, \theta))}{\frac{1}{S} \sum_{s=1}^S \exp(\gamma' g(M^{(s)}, Z_{it}, \theta))}. \quad (6)$$

Solve $(\hat{\theta}, \hat{\gamma})$ jointly by standard GMM on $E[\hat{g}(Z_{it}, \theta, \gamma) \otimes Z_{it}] = 0$. λ is identified the same way it always was in this project — sector–time variation in $\tau\rho_t$, interacted into Z_{it} — since that identification argument was never about *how* the moment is estimated, only about what varies the benefit shifter. $\theta = (\lambda, \delta_0, \delta_1, \delta_2)$ needs at least 4 instrument-interacted moments for point identification (order condition); fewer degrades gracefully to a set estimate rather than failing outright — see the point-vs-set discussion below.

What this buys, relative to MSL and to plain MSM

	recovers f_ψ ?	needs f_ε ?	valid despite nonlinearity?
MSL (headline)	yes	yes	yes (by construction)
plain MSM	no	yes	no — naive-averaging bug
ELVIS	no	no — only the support of M	yes (Thm 2.1)

The real payoff is Remark 2.3: since ρ 's *shape* never enters the estimate, stage 1's deconvolution of f_ε becomes **optional** for stage-2 estimation of θ — only $\hat{\beta}$ is needed to construct $\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}$ in the first place. $\hat{f}_\varepsilon$ can still be reused to build an efficient ρ (less simulation variance for a given S), but that is a computational convenience, not an identification requirement, unlike in MSL where f_ε is load-bearing inside the likelihood itself.

Going further: one unified estimator

If ELVIS handles stage 2's moment on its own, the same device handles the *whole model at once* — because **there is only one true unobservable anywhere in the model**: M_{it} , true materials, for every firm-period, corporation and unincorporated firm alike. Once M_{it} is the ELVIS latent, the production identity and the materials FOC solve in closed form directly — no $\mathcal{V}, \tilde{\mathcal{W}}$ construction, no Φ , no Jacobian, no density-transformation theorem: none of 9999-density-trans.qmd's machinery is needed once M itself, rather than a transformed residual, is what gets integrated against.

This also sidesteps the complication that made the dynamic MSL likelihood “messier” (the 2026-08-12 correction to 9999-density-trans.qmd, replacing bare f_ω with the AR(1) innovation density f_η , recovered by a second Fourier deconvolution): a moment-based estimator only ever needs **moment orthogonality**, $E[\eta_{it}Z_{it}] = 0$ — never the *density* of η .

From M to ε

Start from the materials FOC underlying stage 1 (for any firm, evader or not): $\rho_t M_{it} / (P_t Y_{it}) = \beta e^{-\varepsilon_{it}}$. Solving for ε_{it} directly in terms of the true (latent) M_{it} :

$$\varepsilon_{it}(M_{it}) = \ln \beta - \ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right). \quad (7)$$

This is exactly the stage-1 relation $\ln(\rho_t M_{it}^* / P_t Y_{it}) = \ln \beta - \varepsilon_{it}$, just written with the true M_{it} in place of the observed M_{it}^* — for corporations the two coincide ($M_{it}^* = M_{it}$); for unincorporated firms they don't, and that gap is exactly e_{it} .

A note on measurement. $\rho_t M_{it}$ is a nominal materials expenditure and $P_t Y_{it}$ is nominal sales — both are figures read directly off a firm’s accounts, not quantities separately constructed by dividing a reported value by an independently-estimated price index. Equation 7 only ever uses their *ratio*, so this step never needs ρ_t or P_t individually: it’s immune to whatever price-index quality issue a given dataset has (some surveys report physical quantities and unit prices, allowing a clean deflator; many don’t, forcing a blunter sector-level proxy like a PPI or the GDP deflator). Whenever an *observable* is constructed for the materials-FOC/evasion-FOC side of the model, it should be built from this ratio directly — exactly what `eq-eva-emp` in `600-opt-tax.qmd` already does — not from separately-deflated M_{it}^* and Y_{it} that are then combined.

From M to ω — and why y_{it} cancels

Where the ratio stops being enough. Unlike Equation 7, this step genuinely needs $\ln M_{it}$ and y_{it} *individually* — Cobb-Douglas combines materials with capital and labor, which aren’t measured as a value/price pair the same way, so there’s no ratio to fall back on. This is the one place in the unified system exposed to price-index/deflator quality. That exposure is a known, out-of-scope limitation here (documented, not modeled) — see the corresponding entry in `CLAUDE.md`.

Substitute Equation 7 into the production identity $\omega_{it} = y_{it} - \alpha_K k_{it} - \alpha_L l_{it} - \beta \ln M_{it} - \varepsilon_{it}$:

$$\begin{aligned}\omega_{it}(M_{it}) &= y_{it} - \alpha_K k_{it} - \alpha_L l_{it} - \beta \ln M_{it} - \left[\ln \beta - \ln \rho_t - \ln M_{it} + \ln P_t + \ln Y_{it} \right] \\ &= (y_{it} - \ln Y_{it}) - \alpha_K k_{it} - \alpha_L l_{it} - (\beta - 1) \ln M_{it} - \ln \beta + \ln \rho_t - \ln P_t.\end{aligned}$$

Since $y_{it} \equiv \ln Y_{it}$, the term $(y_{it} - \ln Y_{it})$ is identically zero — y_{it} cancels exactly:

$$\omega_{it}(M_{it}) = (1 - \beta) \ln M_{it} - \alpha_K k_{it} - \alpha_L l_{it} - \ln \beta + \ln \rho_t - \ln P_t. \quad (8)$$

Is this information loss? No — it is Gandhi-Navarro-Rivers’s own $\Phi_{it} = y_{it} - \hat{\varepsilon}_{it}$ construction, specialized to constant (Cobb-Douglas) elasticity, where their Φ loses y_{it} the identical way. The underlying proxy-variable logic (Olley-Pakes/Levinsohn-Petrin/ACF/GNR) is: under the timing assumption, the optimal flexible input is a monotonic function of ω *alone*, so M (equivalently, m) already perfectly proxies ω on its own — y_{it} ’s only informational job in this system is revealing ε_{it} through the (materials-)share equation, which is precisely Equation 7. Once that one job is done, y_{it} has nothing further to say about ω that M_{it} doesn’t already say. y_{it} is used exactly **once** in the whole construction — inside $\varepsilon_{it}(M_{it})$ — not zero times, and not twice.

Caution, not yet resolved. This cancellation is a feature of the *constant-elasticity* (Cobb-Douglas) special case specifically — Equation 8’s algebra depended on $\beta \ln M_{it}$ pulling out linearly. Under a translog production function the map would need rederiving from scratch (flagged in the model notation trap at the top of CLAUDE.md); the qualitative claim that M alone proxies ω under the timing assumption survives, but the closed form here does not automatically generalize.

The AR(1) moment

Raw form, before substituting the FOC (note the coefficients: +1 on ε_{it} , $-\gamma_1$ on ε_{it-1} — not “ $\gamma_1 \varepsilon$ ”):

$$y_{it} - \beta \ln M_{it} - \alpha_K k_{it} - \alpha_L l_{it} - \gamma_0 - \gamma_1 (y_{it-1} - \beta \ln M_{it-1} - \alpha_K k_{it-1} - \alpha_L l_{it-1}) = \eta_{it} + \varepsilon_{it} - \gamma_1 \varepsilon_{it-1}.$$

This is just $y_{it} - \beta \ln M_{it} - \alpha_K k_{it} - \alpha_L l_{it} = \omega_{it} + \varepsilon_{it}$ (production identity) at t and $t-1$, combined with the AR(1) law $\omega_{it} = \gamma_0 + \gamma_1 \omega_{it-1} + \eta_{it}$ — the left-hand side is $(\omega_{it} + \varepsilon_{it}) - \gamma_0 - \gamma_1 (\omega_{it-1} + \varepsilon_{it-1}) = \eta_{it} + \varepsilon_{it} - \gamma_1 \varepsilon_{it-1}$ by substitution.

Since ε_{it} and ε_{it-1} are themselves nothing but Equation 7 evaluated at M_{it} and M_{it-1} , and $\omega_{it}, \omega_{it-1}$ are Equation 8 evaluated the same way, the whole expression reduces with **no ε left anywhere**:

$$\eta_{it}(M_{it}, M_{it-1}) = \omega_{it}(M_{it}) - \gamma_0 - \gamma_1 \omega_{it-1}(M_{it-1}). \quad (9)$$

The moment used is simply $E[\eta_{it}(M_{it}, M_{it-1}) Z_{it}] = 0$ for suitable instruments $Z_{it} = \{k_{it}, l_{it}, \text{lagged } m^*, \dots\}$ — orthogonality only, never a density of η .

Evasion moments, by firm type

Unincorporated firms (interior FOC) — unchanged in substance:

$$e_{it}(M_{it}) = M_{it}^* - M_{it}, \quad \psi_{it}(M_{it}) = h(e_{it}(M_{it})) - \delta_0 + \delta_1 \omega_{it}(M_{it}) - \delta_2 \omega_{it}(M_{it})^2,$$

entering the moment vector as $g = \psi_{it}(M_{it})$, exactly as in Equation 5.

Corporations — relaxed from a hard constraint to a centering moment. The old first-stage device (corp $\Rightarrow e \equiv 0$, imposed *outside* the FOC, purely to build a non-evader sample) becomes a moment corp firms are asked to satisfy, not a restriction used to select them:

$$E[e_{it}(M_{it}) \mid \text{Corp}_i] = 0 \quad \text{or} \quad \text{Med}[e_{it}(M_{it}) \mid \text{Corp}_i] = 0.$$

This matches what the data already show — near-zero, imprecise evasion at both tails — without needing to classify any firm as an exact corner case a priori.

Unincorporated firms — “evade on average,” as a genuine moment inequality. The informal condition “unincorporated firms evade positively on average” is formalized with ELVIS’s own auxiliary-variable device for inequalities (her Example 1.3): introduce a nonnegative auxiliary parameter $U_2 \geq 0$ and impose

$$E[e_{it}(M_{it}) \cdot \mathbb{1}(\text{Unincorp}_i) - U_2] = 0, \quad U_2 \geq 0.$$

Equivalently, an indicator-based version of the same idea — $E[\mathbb{1}\{E[e \mid \text{Unincorp}] > 0\} - 1] = 0$ — pins the *sign* of average evasion for this group to be strictly positive without asserting a specific level; both are moment-inequality constructions, not new machinery beyond what ELVIS already provides for point moments.

Alternative implementation, from Aguiar & Kashaev (2020, *Review of Economic Studies*, “Stochastic Revealed Preferences with Measurement Error,” Lit-Papers/AK2020.pdf) — likely preferred. Their Theorem 4 proves that for a moment system split into inequality moments g_I and equality/centering moments g_M , the γ -components attached to g_I are set-identified only via divergence to $+\infty$ at the optimum — i.e. g_I never actually needs its own γ -search. Their implementation (confirmed by reading their replication code, `TopicsDecisionMaking/ReplicationAK/`) acts on this directly: inequality rows are dropped from the moment vector $\mathbf{g}$ entirely and enforced instead as a **support restriction on the simulator** — their hit-and-run MCMC sampler only proposes draws that already satisfy the model’s inequalities, so `myfun.jl` only ever builds $\hat{g}$ from the equality/centering rows. Applied here: rather than adding U_2 as an auxiliary parameter with its own γ -component, restrict the M -sampler (whatever draws $M^{(s)} \sim \rho(\cdot \mid Z)$ for unincorporated firms) to only propose draws with $e(M^{(s)}) > 0$, and drop the unincorp-evasion-sign row from g altogether. This is a real simplification, not just a computational shortcut: one fewer γ -component to search over, and no risk of the optimizer chasing a $\gamma \rightarrow \infty$ direction (their own stated motivation, since an unbounded γ_0 breaks the standard testing asymptotics — their Assumption 2/Theorem 4 discussion). The U_2 -moment version above is kept as the direct textbook (Example 1.3) construction; see the implementation notes below for the fuller comparison.

The full moment system

Moment	Object	Identifies	Instruments
Materials FOC	$\varepsilon_{it}(M_{it})$	β (mainly via corp)	constant / sector
AR(1) innovation	$\eta_{it}(M_{it}, M_{it-1})$	$\alpha_K, \alpha_L, \gamma_0, \gamma_1$	k_{it}, l_{it} , lagged m^*
Evasion FOC (unincorp)	$\psi_{it}(M_{it})$	$\lambda, \delta_0, \delta_1, \delta_2$	sector-time $\times \tau \rho_t$
Corp evasion	$e_{it}(M_{it})$	pins $M_{it} \approx M_{it}^*$ for corp	— (centering)
Unincorp evasion sign	$e_{it}(M_{it}) \cdot \mathbb{1}(\text{Unincorp})$	tightens the identified set	— (inequality)

All firm-periods — corp and unincorporated — enter jointly; no data is set aside to build a separate first-stage sample. One simulated draw of (M_{it}, M_{it-1}) feeds the entire moment vector for that observation, which is standard for a GMM system with several moments per unit, not double-counting.

Not yet verified — flag, not a result. The algebra above confirms the construction is internally consistent (every moment reduces to a function of M and data alone, with no unobservable left over). It does **not** by itself confirm every parameter in $\theta_{\text{full}} = (\beta, \alpha_K, \alpha_L, \gamma_0, \gamma_1, \lambda, \delta_0, \delta_1, \delta_2)$ is well identified in finite samples — that requires checking the stacked Jacobian $\partial E[g]/\partial \theta_{\text{full}}$ has full column rank numerically, evaluated at a candidate θ , before committing to this as the estimation strategy.

Additional moments from maintained independence assumptions

The table above uses only the moments strictly needed to identify each parameter. But the model maintains several independence assumptions that were never themselves turned into moments — and Schennach’s own supplement ([Lit-Papers/ELVIS_supplement.pdf](#)) gives the formal license to do so. Two results back this up directly:

- **Proposition B.1.** A conditional-mean restriction $E[Y \mid X] = 0$ is equivalent to a *countable* set of unconditional moments $E[Y e^{i\xi X}] = 0$ at rational ξ (an alternative countable basis is in Chamberlain 1987). This is the formal statement behind “interact the residual with more functions of the instruments.”
- **Proposition B.2.** Independence — not just uncorrelatedness — between two unobservable-derived quantities factors the same way through characteristic functions: $E[e^{i\xi X} e^{i\eta Y}] = E[e^{i\xi X}] E[e^{i\eta Y}]$ at rational (ξ, η) . This licenses turning a same-period independence assumption between two structural shocks (e.g. $\varepsilon \perp \psi$) into a moment, not just an uncorrelatedness condition.

In practice, her own examples use a much smaller, practical truncation of this — a mean moment plus a variance/second-moment moment, each with an interacted version. Appendix C’s interval-regression example adds $E[(V^2 - \theta_2)X^2] = 0$ (a homoskedasticity restriction) purely to narrow the identified set; Example 1.5 — her closest analog to this model, a nonlinear errors-in-variables setup — includes a cross-uncorrelatedness moment $E[V_1 V_2] = 0$ between its two error terms *alongside* the FOC-analogue moments, exactly the kind of moment listed below.

Going through every independence assumption this model maintains but hasn’t yet turned into a moment:

Assumption	Where it’s currently used	New moment
$\varepsilon \perp \psi$ (same period)	implicit only, in $f_{\psi, \omega, \varepsilon} = f_{\psi} f_{\omega} f_{\varepsilon}$ (9999-density-trans.qmd)	$E[\varepsilon_{it}(M_{it}) \psi_{it}(M_{it})] = 0$
$\omega \perp \varepsilon$	same factorization	$E[\omega_{it}(M_{it}) \varepsilon_{it}(M_{it})] = 0$
$\psi \perp \omega$	not stated explicitly before	$E[\psi_{it}(M_{it}) \omega_{it}(M_{it})] = 0$ — a genuine overidentifying check: ω already enters ψ ’s definition mechanically through $\delta_1 \omega - \delta_2 \omega^2$, so this tests whether anything is left over once that’s netted out
$\varepsilon_{it} \perp \varepsilon_{it-1}$	flagged as new-but-maintained in 9999-density-trans.qmd’s dynamic extension, never a moment	$E[\varepsilon_{it}(M_{it}) \varepsilon_{it-1}(M_{it-1})] = 0$
$\varepsilon_{it-1} \perp \psi_{it}$	same	$E[\varepsilon_{it-1}(M_{it-1}) \psi_{it}(M_{it})] = 0$
$\varepsilon \perp Z$	mean version already in the materials-FOC row	add: $E[\varepsilon_{it}(M_{it})^2 - \sigma_{\varepsilon}^2] = 0$ and $E[(\varepsilon_{it}(M_{it})^2 - \sigma_{\varepsilon}^2) Z_{it}] = 0$
$\psi \perp Z$	mean version already in the evasion-FOC row	add: $E[\psi_{it}(M_{it})^2 - \sigma_{\psi}^2] = 0$ and $E[(\psi_{it}(M_{it})^2 - \sigma_{\psi}^2) Z_{it}] = 0$

The lag moments come essentially for free: (M_{it}, M_{it-1}) are already jointly drawn for the AR(1) row, so these are just extra rows in g evaluated on a draw that’s already being simulated. **Not a new moment:** $M \perp e \mid \omega$ — that’s the structural reason $e(M)$ can be written as a function of M alone in the first place (§ “The one unobservable” above); it’s already fully used by construction, not an additional testable restriction.

This isn’t free. Theorem F.1 in the supplement gives a closed-form confidence-region critical value of $\chi_{d_g, \alpha}^2$, where d_g is the number of moments — every row added raises the bar $-2n\hat{L}_n(\theta)$ has to clear. More moments can tighten the identified set (that’s the whole point, per Appendix C’s examples), but it’s not automatic, and it isn’t free computationally either: her own timing note for these simulation-based estimates ranges from “a few minutes... to a few hours (for the one with 27 moment conditions).” Recommended sequencing: implement the mean-only cross moments above first (six cheap extra rows), and treat the full characteristic-function grid implied by Prop. B.2 as a later robustness/tightening pass rather than a day-one requirement.

Implementation notes from the reference code

Lit-Papers/ELVIS_code/ (elvisutil.g, elvisexample.g) is Schennach’s own GAUSS implementation, referenced from the supplement. A few details worth carrying into any implementation here, beyond what’s stated in the main text above:

- **Objective function.** Definition F.1 in the supplement covers a whole class of objectives; two concrete choices are singled out — empirical likelihood (Owen), and a continuously-updated-GMM-type quadratic $\hat{L}_n^{\text{CUE}}(\theta) = \sup_{\gamma} -\frac{1}{2}\hat{g}'(\theta, \gamma)\hat{V}^{-1}(\theta, \gamma)\hat{g}(\theta, \gamma)$ with $\hat{V}(\theta, \gamma) = \frac{1}{n} \sum_i \tilde{g}(Z_i, \theta, \gamma)\tilde{g}(Z_i, \theta, \gamma)'$. The reference code defaults to the CUE version (`#define DOGMM`), with EL available as a toggle.
- **Sampling from the tilt directly.** Rather than computing Equation 6 as a self-normalized importance-sampling ratio over draws from the *untilted* ρ , the reference code runs a Metropolis chain that samples directly from the γ -tilted density $\rho(u | z) \exp(\gamma'g(u, z, \theta))$ (with equilibration/burn-in steps discarded) and simply averages g over the retained draws. This is mathematically equivalent to Equation 6 but numerically more robust — importance-sampling ratios degrade badly when the tilt moves far from ρ , which is plausible here given γ enters nonlinearly and the support of M is bounded ($M \in (M_{it}^* - 1/2\lambda, M_{it}^*)$).
- **What gets optimized how.** θ ’s own optimization problem is “difficult” — it may be non-convex and reached over a whole set rather than a point (exactly the set-identification possibility this model already anticipates) — so the reference implementation scans θ over a grid rather than searching for it directly. For each grid point, (nuisance, γ) — a well-behaved, convex problem with a unique global optimum for fixed θ — is optimized via a derivative-free simplex method (Nelder-Mead), avoiding any need for analytic derivatives of g . This matches the “coordinate-wise profiling” approach already planned for θ_{full} ’s many parameters.
- **Nuisance parameters as a first-class device.** The code explicitly separates the full parameter vector into parameters of interest and nuisance parameters that get profiled out jointly with γ (rather than gridded over) — exactly the role the auxiliary revenue parameter $\tilde{\theta}(\tilde{\tau})$ plays below.

A second reference implementation: Aguiar & Kashaev (2020)

Beyond Schennach’s own GAUSS code, a full applied ELVIS implementation exists in economics: Aguiar & Kashaev (2020, *Review of Economic Studies*, “Stochastic Revealed Preferences with Measurement Error,” [Lit-Papers/AK2020.pdf](#)) build an ELVIS-based test of utility-maximization/revealed-preference rationality under measurement error in consumption panel data — their moment system mixes revealed-preference *inequality* moments g_I (analogous to our unincorp evasion-sign moment) with measurement-error *centering* moments g_M (analogous to our corp centering / materials-FOC / AR(1) rows), exactly our equality/inequality split. Their replication code (Julia + CUDA, [TopicsDecisionMaking/ReplicationAK/](#)) is a working, GPU-accelerated ELVIS estimator, not just theory — worth consulting directly if a from-scratch implementation is built here. Details confirmed by reading the code itself:

- **Theorem 4 (their semi-analytic reduction) is implemented literally, not just proved.** g_I never enters the moment vector `gvec/h_M` that gets γ -searched (`myfun.jl`). Instead their simulator — a hit-and-run MCMC chain — is restricted to only propose draws already satisfying the inequalities; the γ -search happens purely over the equality/centering block. See the alternative unincorp-inequality implementation noted above, which follows this pattern directly.
- **Objective.** Same CUE-type quadratic already noted from the GAUSS code ($\hat{L}_n^{\text{CUE}}$), built as $\frac{1}{2}\hat{g}'\hat{\Omega}^-\hat{g}$ with $\hat{\Omega}^-$ a generalized inverse obtained by eigendecomposing the sample covariance of the tilted moments and dropping non-positive eigenvalues (`cuda_fastoptim.jl`, function `objMCcu`) — a concrete, numerically-robust recipe for the pseudo-inverse Definition F.1 leaves unspecified.
- **Solver.** Not Nelder-Mead (unlike the GAUSS reference code) — they use derivative-free, bound-constrained **BOBYQA** (via `NLopt`) for the γ -search, warm-started from a global/derivative-free pre-search (`BlackBoxOptim`), then run twice as a cheap restart-based convergence check.
- **Simulation.** They implement the direct Metropolis-sampling-from-the-tilted-density approach (matching the “sampling from the tilt directly” bullet above), and critically generate the constraint-satisfying MCMC chain **once** and reuse it — unchanged — across every candidate γ (and, if gridding, every candidate θ). Chain generation is the expensive, one-time cost (their timing: hours); each subsequent γ -evaluation is cheap since it only resamples/reweights the cached chain. Directly reusable here: generate one M -chain (or one chain per firm-period) satisfying the feasibility support $\mathcal{U}_{Z,\theta} = (M_{it}^* - 1/(2\lambda), M_{it}^*)$ once per θ -grid-point, not once per γ -evaluation.
- **Per-unit warm start.** Before the chain runs, they solve a genuinely convex per-unit QP (`Convex.jl` + `ECOS`) minimizing squared measurement error subject to the model’s linear feasibility constraints, to get a feasible starting draw for each unit’s chain — this is where a “unique global minimum” property is actually exploited, at the per-unit warm-start stage, not in the γ -search itself (their own footnote is easy to misread as being about the γ -objective).

The counterfactual without f_ψ

MSL’s expected-revenue integral at a candidate tax rate $\tilde{\tau}$ weights firm outcomes by $f_\psi(\psi(\varepsilon)) J_i(\varepsilon) f_\omega(\omega(\varepsilon)) f_\varepsilon(\varepsilon)$ — the full recovered density stack. ELVIS never recovers f_ψ , so that integral is not directly available. Schennach’s framework has its own device for exactly this situation — bounding a functional of the unobservable that is not itself one of the estimating moments. **Citation correction:** this worked example is not in the 2014 *Econometrica* paper itself (that paper really does run pp. 345–385, as noted when double-checking this — no p. 1250 there). It’s in her 2022 *Journal of Economic Literature* survey, “Measurement Systems” (Lit-Papers/schennach-2022-measurement-systems.pdf; bib key Schennach2021, the working-paper vintage — page numbers below are from the published JEL Vol. LX, December 2022 version), p. 1250, immediately after she introduces the ELVIS moment $\tilde{g}(z, \theta, \gamma)$ (her eq. 68, our Equation 2) and Theorem 11 (her version of Theorem 2.1 above):

“One should be careful not to interpret the entropy maximizing distribution as the actual distribution of the unobservables. Nevertheless, one can easily bound features of the true distribution of U , provided that these features can be expressed as moment conditions. For instance, if one wished to find all possible values of the mean of U consistent with the data, one could merely include $E[U - \tilde{\theta}] = 0$ among the moment conditions and the identified set for the auxiliary parameter $\tilde{\theta}$ will indicate the possible values of $E[U]$.”

That’s the auxiliary-parameter device used below, in exactly this form — an extra moment $E[U - \tilde{\theta}] = 0$ appended to the system, with the *identified set for $\tilde{\theta}$* (not a point estimate) read off as the answer — just with $U = R_i(\tilde{\tau}; M_i)$ (counterfactual revenue) here instead of a generic U .

A tempting shortcut — reuse the entropy-maximizing tilt μ^* from Equation 4, evaluated at $\hat{\theta}, \hat{\gamma}$, as if it were “the” posterior over M_{it} , and average the counterfactual outcome against it — would give a number, but understates uncertainty: μ^* is only the entropy-maximizing member of the (possibly large) set of distributions consistent with the estimated moments, not necessarily the true one. Schennach is explicit about this. The honest version instead re-solves for the functional’s own identified set, using the same machinery, rather than trusting one particular tilt.

The auxiliary-moment construction

Hold $\hat{\theta}$ fixed — the counterfactual is a forward simulation given already-estimated parameters, not new estimation. Reuse the closed-form counterfactual evasion FOC (unchanged in derivation from the MSL version, since it only used the linear- q FOC, not any density):

$$\tilde{e}_i(\tilde{\tau}) = \tilde{e}(\tilde{\tau}; \omega_i(M_i), \psi_i(M_i)) = \frac{\tilde{\tau} \rho_t - C(\omega_i, \psi_i)}{2\lambda \tilde{\tau} \rho_t}, \quad C(\omega, \psi) \equiv \mathbf{e}^{\delta_0 - \delta_1 \omega + \delta_2 \omega^2 + \psi},$$

with the true M_i itself unchanged under the counterfactual (the materials FOC doesn't depend on τ), and form firm i 's counterfactual revenue directly as a function of the same latent M_i :

$$R_i(\tilde{\tau}; M_i) = \tilde{\tau} \left[P_t Y_{it} - \rho_t M_i - (1 - q(\tilde{e}_i(\tilde{\tau}))) \rho_t \tilde{e}_i(\tilde{\tau}) \right].$$

For a candidate $\tilde{\tau}$, introduce an auxiliary parameter $\tilde{\theta}(\tilde{\tau}) \equiv \frac{1}{N} \sum_i E[R_i(\tilde{\tau}) | Z_i]$ and append its own moment to the system:

$$E[R_i(\tilde{\tau}; M_i) - \tilde{\theta}] = 0,$$

with θ fixed at $\hat{\theta}$ — only $\tilde{\theta}$ and an extended γ (one more component, for the new moment) are free. The estimator is the same simulated-draw, ratio-of-averages machinery as Equation 6, just evaluated on the extended moment vector including this new row.

Point or set — decided by the procedure, not assumed

For a fixed $\tilde{\tau}$, grid over **candidate revenue values** $\tilde{\theta}$ (not over the structural θ — that part is already fixed at $\hat{\theta}$):

1. For each candidate $\tilde{\theta}$, solve the inner problem — maximize the chosen objective (EL or the CUE-type quadratic, per Definition F.1 in the supplement) over the extended γ , with the row $R_i(\tilde{\tau}; M_i) - \tilde{\theta}$ included in g .
2. That maximized value is a criterion function $\hat{L}_n(\tilde{\theta}; \tilde{\tau})$ — the identical construction used to estimate θ itself, just evaluated on a grid over $\tilde{\theta}$ instead of solved for a point.
3. **Profile it against the unrestricted model, don't use it raw.** The correctly-sized statistic subtracts the global maximum over the whole system, $\hat{Q}_n(\tilde{\theta}; \tilde{\tau}) = -(\hat{L}_n(\tilde{\theta}; \tilde{\tau}) - \sup_{\theta', \gamma} \hat{L}_n(\theta', \gamma))$ — not $\hat{L}_n(\tilde{\theta}; \tilde{\tau})$ against a threshold on its own. Subtracting the unrestricted maximum is what gives the efficiency/degrees-of-freedom improvement this construction is designed for (Chernozhukov-Hong-Tamer 2007's own convention).
4. The identified/confidence set for expected revenue at $\tilde{\tau}$ is $\{\tilde{\theta} : n\hat{Q}_n(\tilde{\theta}; \tilde{\tau}) \leq c_\alpha\}$. Two ways to get c_α : subsampling per Chernozhukov, Hong & Tamer (2007) for a tight region, or — for a fast first pass — Theorem F.1's ready closed form, $c_\alpha = \chi_{d_g, \alpha}^2$ (the $(1 - \alpha)$ quantile of a chi-squared with d_g = number of moments in the extended system), valid whenever the objective's weighting is the efficient one ($W = \hat{V}^-$). The χ^2 shortcut is more conservative — it skips the profiling step in point 3 — but needs no subsampling and no resampling loop, just a table lookup.

Repeating this across a grid of $\tilde{\tau}$ produces a band, not a single curve, at each candidate tax rate — and the **shape decides the regime**: a sharp trough in $\hat{Q}_n(\cdot; \tilde{\tau})$ at every $\tilde{\tau}$ collapses the band to a single Laffer curve (point identification); a flat plateau at some $\tilde{\tau}$ leaves a genuine

interval there — the model does not pin expected revenue down more tightly at that rate, and no amount of extra machinery changes that; only more/better instruments would. This is why the point-vs-set question is never decided *ex ante*: the same grid-and-criterion-function procedure produces both regimes automatically, and the trough width is read off after running it. $\tilde{\tau}^*$ then follows by reading off where the (possibly wide) revenue band peaks — a range of candidate optima if the argmax band itself is wide, a single point if it is not.

Convergence-rate caveat. Which regime shows up also affects *how fast* the region shrinks with sample size. Per Appendix F.2 of the supplement, confidence regions of this form are root- n consistent whenever the objective is exactly flat over (an asymptotic neighborhood of) the identified set — the “degeneracy” property, satisfied in her interval-data and censored-regression examples — but only $\sqrt{\ln n/n}$ -consistent otherwise, which is the case for some of her own measurement-error extensions of the errors-in-variables example closest to this model. Whether our system is degenerate isn’t something to assume either way; it’s a property of the fitted $\hat{Q}_n$ to check once real estimates exist, not before.

Summary

- ELVIS closes the exact gap the naive-averaging GMM attempt fell into: it derives, rather than assumes, the missing posterior reweighting, as the entropy-maximizing Lagrange multiplier on the moment constraint (Theorem 2.1).
- Applied to stage 2 alone (recasting ψ ’s moment in M_{it} instead of ε_{it}), it delivers $\theta = (\lambda, \delta_0, \delta_1, \delta_2)$ without f_ψ (same trade-off MSM tried to make, but valid) and — via Remark 2.3 — without even needing f_ε , only $\hat{\beta}$.
- Applied to the whole model at once, the same latent M_{it} absorbs *everything*: $\beta, \alpha_K, \alpha_L, \gamma_0, \gamma_1$ (production side, corp and unincorp jointly) and $\lambda, \delta_0, \delta_1, \delta_2$ (evasion side), in one moment system, with no separate first stage and no density-transformation machinery at all.
- The counterfactual survives without f_ψ by treating expected revenue as its own auxiliary-parameter moment, solved with the same entropic-tilting machinery — and the point-vs-set question is resolved by the shape of the resulting (properly profiled) criterion function, not assumed in advance.
- Every observable used for the materials-FOC/evasion-FOC moments ($\mathcal{V}_{it}, \varepsilon_{it}(M_{it})$) is built from the *nominal* materials-to-sales ratio directly — never from separately-deflated M_{it}^*, Y_{it} — so that side of the system is immune to price-index measurement quality; only $\omega(M)$ and the AR(1) moment, which need $\ln M_{it}$ and y_{it} individually, inherit that exposure (documented, not modeled — see CLAUDE.md).
- Beyond the baseline moment table, every independence assumption already maintained in the model (, , , $_it$ $_it-1$, $_it-1$ $_it$, plus variance-matching versions of Z and Z) converts into an additional, essentially-free moment via Propositions B.1–B.2 in the

supplement — more identifying power, at the cost of a higher ² critical-value threshold and more simulation compute per added row.

- A second, working economics implementation of ELVIS exists — Aguiar & Kashaev (2020, *Review of Economic Studies*), [Lit-Papers/AK2020.pdf](#), code in [TopicsDecisionMaking/ReplicationAK/](#) — confirming in practice (not just theory) that inequality moments can be dropped from the γ -searched vector entirely and enforced instead as a support restriction on the simulator (their Theorem 4); applied above to drop the unincorp evasion-sign moment in favor of restricting the M -sampler. Their code also gives concrete, reusable numerical choices: BOBYQA (not Nelder-Mead) for the γ -search, an eigenvalue-truncated generalized inverse for $\hat{\Omega}^-$, direct tilted-density sampling with the constraint-satisfying chain cached once and reused across every γ (and θ -grid point), and a convex per-unit QP warm start.
- **Open, not yet done:** numerically verify the stacked Jacobian’s full column rank for θ_{full} ; work out an efficient default ρ (reusing $\hat{f}_\varepsilon$, translated to a distribution over M); confirm the translog case (if ever pursued) needs its own version of Equation 8; check whether the fitted system is degenerate in Chernozhukov-Hong-Tamer’s sense, since that governs whether inference converges at $\sqrt{n}$ or only $\sqrt{\ln n/n}$.